# REPRESENTED BY

Name : Nikunj Kumar

Branch : Computer Science and Engineering

Registration No. : 24105141052

# Introduction

- A calculator is a software application used to perform mathematical calculations.
- This project is developed using Java Swing to create a graphical user interface that allows users to perform different arithmetic operations easily.

# Objectives

- Develop a GUI calculator using Java
- Perform basic arithmetic operations
- Understand event handling in Java Swing
- Implement additional features like currency conversion

# Technologies Used

- Java Programming Language
- Java Swing (GUI Library)
- AWT Event Handling
- Object Oriented Programming (OOP)

# Features of the Calculator

- Addition (+)
- Subtraction (−)
- Multiplication (×)
- Division (÷)
- Percentage (%)
- Square, Square Root
- Reciprocal (1/x)

# Program Logic

```
double calculate(double a, double b, String op) {

    switch(op) {
        case "+": return a + b;
        case "-": return a - b;
        case "*": return a * b;
        case "/": return (b != 0) ? a / b : 0;
        double square(double x){ return x * x; }

       double sqrt(double x){ return Math.sqrt(x); }

       double percentage(double x){ return x / 100; }

       double reciprocal(double x){ return 1 / x; }
    }

       return 0;
}
```

# Working of the Program

Start

↓

Enter Number

↓

Select Operator

↓

Enter Second Number

↓

Press "="

↓

Perform Calculation

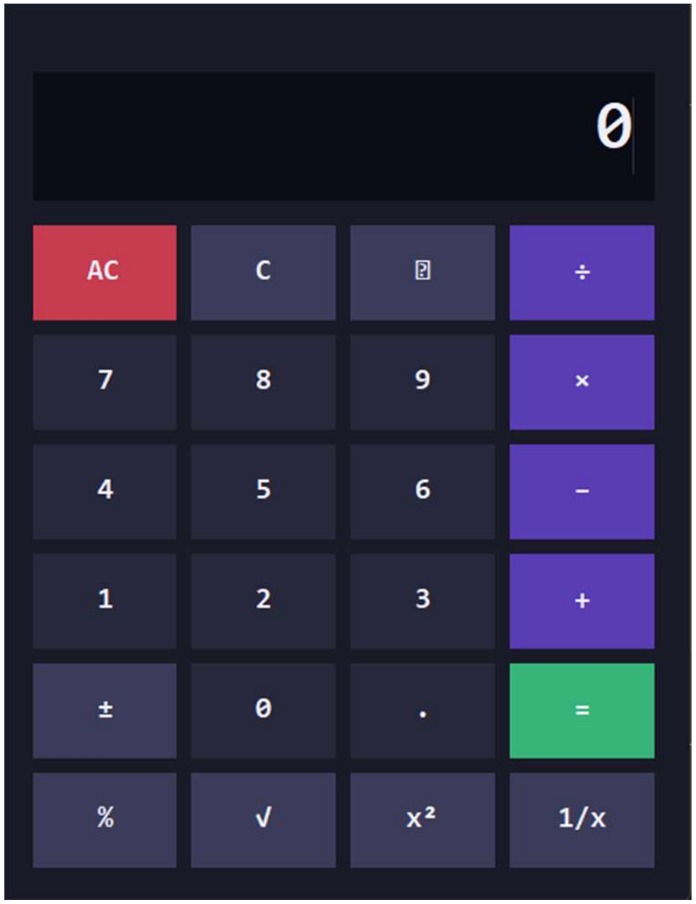
↓

Display Result

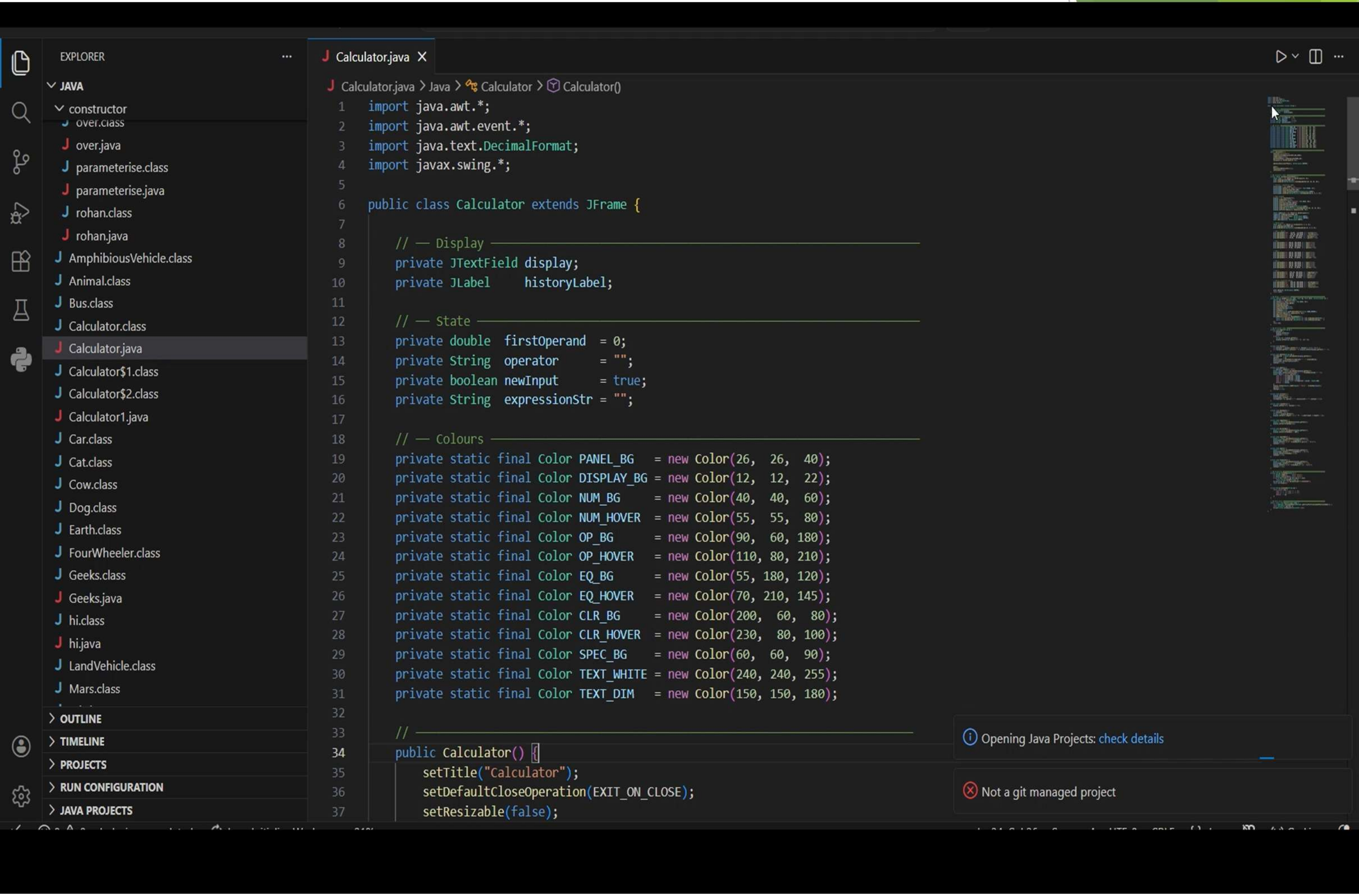
↓

End

# Calculator Interface (GUI)



# Output


EXPLORER
JAVA
constructor
over.class
over.java
parameterise.class
parameterise.java
rohan.class
rohan.java
AmphibiousVehicle.class
Animal.class
Bus.class
Calculator.class
Calculator.java
Calculator$1.class
Calculator$2.class
Calculator1.java
Car.class
Cat.class
Cow.class
Dog.class
Earth.class
FourWheeler.class
Geeks.class
Geeks.java
hi.class
hi.java
LandVehicle.class
Mars.class
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
Calculator.java
Calculator.java > Java > Calculator > Calculator()
import java.awt.*;
import java.awt.event.*;
import java.text.DecimalFormat;
import javax.swing.*;
public class Calculator extends JFrame {
// — Display
private JTextField display;
private JLabel historyLabel;
// — State
private double firstOperand = 0;
private String operator = "";
private boolean newInput = true;
private String expressionStr = "";
// — Colours
private static final Color PANEL_BG = new Color(26, 26, 40);
private static final Color DISPLAY_BG = new Color(12, 12, 22);
private static final Color NUM_BG = new Color(40, 40, 60);
private static final Color NUM_HOVER = new Color(55, 55, 80);
private static final Color OP_BG = new Color(90, 60, 180);
private static final Color OP_HOVER = new Color(110, 80, 210);
private static final Color EQ_BG = new Color(55, 180, 120);
private static final Color EQ_HOVER = new Color(70, 210, 145);
private static final Color CLR_BG = new Color(200, 60, 80);
private static final Color CLR_HOVER = new Color(230, 80, 100);
private static final Color SPEC_BG = new Color(60, 60, 90);
private static final Color TEXT_WHITE = new Color(240, 240, 255);
private static final Color TEXT_DIM = new Color(150, 150, 180);
//
public Calculator() {
setTitle("Calculator");
setDefaultCloseOperation(EXIT_ON_CLOSE);
setResizable(false);
Opening Java Projects: check details
Not a git managed project

# Advantages

- Easy to use interface
- Fast calculations
- Multiple operations available

# Conclusion

- The Java Calculator project demonstrates the use of Java Swing for GUI development and event handling to perform mathematical operations efficiently.